

PROJECT – STRATEGIC UNIVERSITY MANAGEMENT AND DECISION SUPPORT SYSTEM

1. Project Objective

The primary objective of the system is to provide university senior management with an integrated strategic decision support platform that enables them to:

- Monitor the overall status and performance of the university from a holistic perspective.
- Compare the performance of faculties and academic departments.
- Evaluate financial, academic, and physical resources.
- Track the progress of strategic goals and institutional objectives.
- Analyze the potential outcomes of alternative strategic decisions.
- Identify institutional risks at an early stage.
- Support medium- and long-term decision-making through data-driven analyses.
- Monitor institutional compliance with and improvement toward **THE (Times Higher Education), QS, and the Turkish Council of Higher Education (YÖK)** university evaluation and ranking criteria.

The proposed system is not intended to replace operational systems, such as the Student Information System (SIS), Human Resources Management System (HRMS), or Financial Management System. Instead, it will integrate data collected from these systems to provide senior management with consolidated analyses, performance indicators, and strategic decision alternatives.

2. Core Data Model and University Structure

The system shall support the definition and management of the following core university entities:

- Faculties, academic departments, graduate schools, and vocational schools (initially, the Faculty of Engineering and Architecture may be considered as the pilot unit).
- Administrative and support units (e.g., Student Affairs Office, Erasmus Office, etc.).
- Students.
- Academic and administrative staff.
- Academic programs, curricula, and courses.
- Classrooms, laboratories, and offices.
- Libraries and shared facilities.
- Revenue and expenditure categories.
- The university's strategic goals and objectives.

The majority of these data will be obtained from the university's existing information systems and transformed into an analyzable format within the decision support platform.

3. Strategic Education and Student Analytics

The system shall be capable of analyzing the following educational and student-related performance indicators:

- Total number of students at both university-wide and departmental levels.
- Number of newly enrolled, active, and graduated students.
- Program quotas and enrollment (occupancy) rates.
- Student academic performance and graduation rates.
- Number of preparatory school students.
- Average time to graduation.
- Student attrition and non-renewal rates.
- Percentage of scholarship students.
- Percentage of international students.
- Trends in student demand for academic programs, including minimum admission score analyses at both the Ankara and national levels.
- Comparative analyses of student enrollment across similar universities and academic departments.

These analyses will support the evaluation of academic program effectiveness and provide insights into future student demand.

4. Academic Staff Performance Analysis

The system shall evaluate academic staff performance across the following key dimensions:

- Teaching activities and course workload.
- Academic staff performance evaluation results.
- Student supervision and graduate thesis advising.
- Scientific publications and citation performance.
- National and international research projects.
- Patents, innovation, and commercialization activities.
- Administrative responsibilities.
- University–industry collaboration.
- Community engagement and societal contribution.
- National and international recognition.

The system shall support performance comparisons based on:

- Individual academic staff members,
- Academic departments,
- Faculties,
- Academic titles/ranks,
- Academic years.

In addition, trend analyses shall be provided across different time periods. The weighting of performance indicators shall be configurable according to the university's strategic priorities and may vary among academic departments.

5. Physical Resources and Capacity Analysis

The system shall evaluate the utilization and capacity of the university's physical resources, including:

- Classrooms.
- Laboratories.
- Academic and administrative offices.
- Library facilities.
- Meeting and conference rooms.
- Social and shared-use spaces.

The following analyses shall be supported:

- Classroom occupancy rates.
- Laboratory utilization rates.
- Space allocation by organizational unit.
- Space available per student and staff member.
- Identification of underutilized or overcrowded facilities.
- Forecasting future requirements for buildings, classrooms, and laboratories.
- Assessing the impact of new academic programs on existing physical capacity.

6. Strategic Financial Analysis

Revenue Sources

The system shall support the analysis of the following revenue categories:

- Tuition fees
- Research project revenues
- Certificate and continuing education program revenues
- University–industry collaboration revenues
- Donations (optional)
- Sponsorship revenues (optional)
- Other operational revenues

Expenditure Categories

The system shall support the analysis of the following expenditure categories:

- Academic staff salaries
- Administrative staff salaries
- Educational expenses
- Research and development (R&D) expenditures
- Scholarship expenditures
- Building maintenance and energy costs

- Technology and laboratory expenses
- Central administrative expenses
- Capital investment expenditures

The system shall calculate the following strategic financial performance indicators:

- Overall revenue–expenditure balance
- Financial status by faculty and academic department
- Revenue per student
- Cost per student
- Cost per graduate
- Ratio of personnel expenses to total expenditures
- Impact of scholarship expenditures on total revenue
- Percentage of research income within total revenue
- Budget target achievement rate for each organizational unit (optional)

7. Academic Program Sustainability Analysis

Each academic program shall be evaluated using a multidimensional framework rather than solely on financial performance.

The evaluation criteria shall include:

- Student demand
- Program enrollment (occupancy) rate
- Graduation rate
- Graduate employability and career success
- Academic staff qualifications and educational quality
- Research performance
- Revenue–expenditure balance
- Physical resource requirements
- Contribution to the university's strategic objectives
- Contribution to regional development and societal needs
- Contribution to the university's institutional reputation and brand value

Based on the analysis, academic programs shall be classified into categories such as:

- Programs requiring strengthening
- Programs with growth potential
- Programs requiring restructuring
- Programs suitable for possible consolidation or merger
- Programs requiring strategic institutional support

8. University Performance Management and Monitoring

The overall institutional performance of the university shall be monitored across the following strategic dimensions:

1. Education and teaching quality (including course and institutional evaluation survey results)
2. Research and development (e.g., project scale, complexity, scientific impact)
3. Financial sustainability
4. Human resources (e.g., recruitment, applications, turnover rates)
5. Internationalization (e.g., international partnerships and collaborations)
6. Community engagement and social impact (e.g., public lectures, outreach programs, weekend workshops for gifted students)
7. Student and graduate success
8. University–industry collaboration
9. Quality assurance and accreditation
10. Physical and technological infrastructure
11. Institutional reputation and visibility

For each Key Performance Indicator (KPI), the system shall display:

- Current value
- Target value
- Achievement rate
- Previous year's value
- University-wide average
- Faculty and departmental comparisons
- Risk level and recommended corrective actions

The system shall enable senior management to identify which strategic objectives are progressing as planned, which are delayed, and which are at risk.

9. THE, QS, and YÖK Performance Monitoring Module

The objective of this module is to centrally collect, validate, and monitor the university's performance data related to the **Times Higher Education (THE)**, **QS World University Rankings**, and the **Turkish Council of Higher Education (YÖK) University Monitoring and Evaluation System**, while providing strategic recommendations for institutional improvement.

THE Performance Indicators

The module shall support the following THE assessment dimensions:

- Teaching Environment
- Research Environment
- Research Quality
- International Outlook
- Industry Income and Patents

The **THE 2026 methodology** evaluates universities using **18 performance indicators** grouped under these five strategic dimensions.

QS Performance Indicators

The module shall support the following QS assessment dimensions:

- Academic Reputation
- Citations per Faculty Member
- Employer Reputation
- Graduate Employment Outcomes
- Faculty-to-Student Ratio
- International Faculty Ratio
- International Student Ratio
- International Research Network
- Sustainability

The QS methodology evaluates institutional performance across the areas of research, employability, learning experience, global engagement, and sustainability.

YÖK Performance Indicators

The module shall support the following YÖK assessment dimensions:

- Education and Teaching
- Research, Development, Projects, and Publications
- Internationalization
- Sustainability
- Community Service and Social Responsibility

The YÖK University Monitoring and Evaluation System assesses universities based on these five principal categories.

University data shall be mapped to the corresponding indicators of THE, QS, and YÖK wherever applicable. The system shall generate an independent data readiness and compliance score for each evaluation framework.

Example

- THE Data Readiness Score: **65%**
- QS Data Readiness Score: **52%**
- YÖK Data Readiness Score: **64%**

Benchmarking and Comparative Analysis

The system shall support benchmarking against:

- Previous years' institutional performance
- National averages within Türkiye
- Comparable universities
- Selected competitor universities

What-if Scenario Analysis

The module shall support strategic simulation scenarios such as:

- How would increases in citation counts affect THE and QS rankings?
- How would an increase in academic staff influence the student-to-faculty ratio?
- Which performance indicators would improve if the proportion of international students increased?
- How would increasing the number of doctoral graduates affect THE and YÖK performance metrics?
- Which ranking criteria would benefit from increased research funding and higher university–industry collaboration revenues?

10. Executive-Level Strategic Scenarios

The system shall support the comparison of the potential outcomes of different strategic decision alternatives.

10.1. Student Enrollment Scenario

The system shall analyze questions such as:

- How would an increase or decrease in student enrollment affect total revenue?
- How would academic staffing requirements change?
- Would classroom and laboratory capacity remain sufficient?
- How would the cost per student change?

10.2. Tuition Fee and Scholarship Scenario

The system shall analyze questions such as:

- What would be the impact of changes in tuition fees on total revenue?
- How would an increase in scholarship rates affect the institutional budget?
- How might different tuition fee and scholarship policies influence student demand?

10.3. New Academic Program Scenario

The system shall evaluate:

- The investment required to establish a new academic program
- The number of academic staff members required

- The minimum number of students needed to reach the break-even point
- The adequacy of existing physical capacity
- The strategic contribution of the proposed program

10.4. Academic Program Restructuring Scenario

The system shall analyze:

- The potential benefits of merging programs with low student demand
- The effects of closing a program on revenue, expenditure, and resource utilization
- The impact of such decisions on students, staff, and institutional reputation

10.5. Academic Staffing Scenario

The system shall evaluate:

- How educational and research indicators would change if new academic staff members were recruited
- The impact of salary increases on the institutional budget
- The relationship between the number of academic staff members and additional teaching payments
- How changes in course enrollment limits would affect educational quality and cost
- How recitation and question-and-answer sessions delivered by research assistants and students would affect quality and cost

10.6. Investment Scenario

The system shall analyze:

- The cost of investing in new buildings, laboratories, or technological infrastructure
- The contribution of the investment to utilization capacity and academic performance
- The expected return on investment period

10.7. Research Strategy Scenario

The system shall evaluate:

- How increased research incentives could affect publication and project outputs
- How increased project revenues would influence financial sustainability
- The institutional impact of prioritizing selected research areas

10.8. Economic Risk Scenario

The system shall analyze:

- How rising inflation would affect institutional expenditures

- How exchange-rate fluctuations would influence technology and laboratory costs
- Which measures could be implemented in the event of a decline in student-related revenues

Senior management shall be able to define additional scenarios according to institutional needs. A dedicated **Scenario Definition and Simulation Cockpit** shall be developed for this purpose.

11. Risk and Early Warning System

The system shall generate alerts for senior management when any of the following conditions occur:

- Program enrollment rates fall below a critical threshold
- Student attrition rates increase
- A budget deficit emerges
- Expenditures of an organizational unit exceed its allocated budget
- The need for additional academic staff increases
- Classroom or laboratory capacity becomes insufficient
- Academic performance indicators decline
- Strategic objectives are delayed
- Accreditation renewal or expiration dates are approaching
- Research revenues or student-related revenues decrease

12. Executive Dashboard

The executive management interface shall provide a consolidated overview of the following indicators:

- Total number of students and enrollment trends
- Academic program enrollment rates
- Graduation and student attrition rates
- Total revenue and expenditure
- Budget realization rate
- Cost per student
- Sustainability status of academic programs
- Distribution of academic staff and teaching workload
- Publication, citation, and research project performance
- Scholarship costs
- Physical capacity utilization rates
- Progress toward strategic objectives
- Critical risks and early-warning notifications
- THE, QS, and YÖK data readiness scores
- Strong and weak performance indicators

The dashboard shall support both **drill-down** and **roll-up** navigation, enabling users to move from summary-level information to detailed data and from detailed data back to consolidated institutional views.

13. Core System Modules

The system may consist of the following main modules:

1. University Structure and Core Data Management
2. Strategic Education and Student Analytics
3. Academic Staff and Performance Analytics
4. Physical Resource and Capacity Analysis
5. Financial Sustainability Analysis
6. Academic Program Sustainability Analysis
7. University Performance Management
8. Strategic Plan and Objective Monitoring
9. What-if Scenario Analysis
10. THE, QS, and YÖK Evaluation and Monitoring Management
11. Risk and Early Warning Management
12. Executive Dashboard and Strategic Reporting
13. Data Integration
14. User and Authorization Management